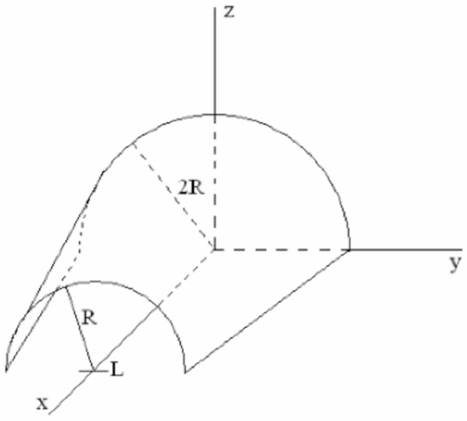
ECE 329 **Homework 5** **Due: Thursday, October 1, 2026, 11:59:00PM**

- Homeworks are due Thursday at 11:59:00 p.m. Late homework will not be accepted.
- Homeworks are to be turned in to Gradescope.
- Any deviation from the following steps will result in a 5 point penalty:
  - Write your name, netID, and section ONLY on the first page of your submission.
  - Start each new problem on a new page.
  - Scan the homework as a PDF (rather than taking pictures). Use a free scanning app if you do not have access to a photocopier.
  - Upload the PDF to Gradescope and tag each problem's location in the PDF.
- Each student must submit individual solutions for each homework. You may discuss homework problems with other students registered in the course, but you may not copy their solutions. If you use any source outside of class materials that we've provided, you must cite every source that you used.
- Use of homework solutions from past semesters is not allowed and is considered cheating. Copying homework solutions from another student or generative AI is considered cheating.
- Penalties for cheating on homework: a zero for the assignment on the first offense, and an F in the course on the second offense.

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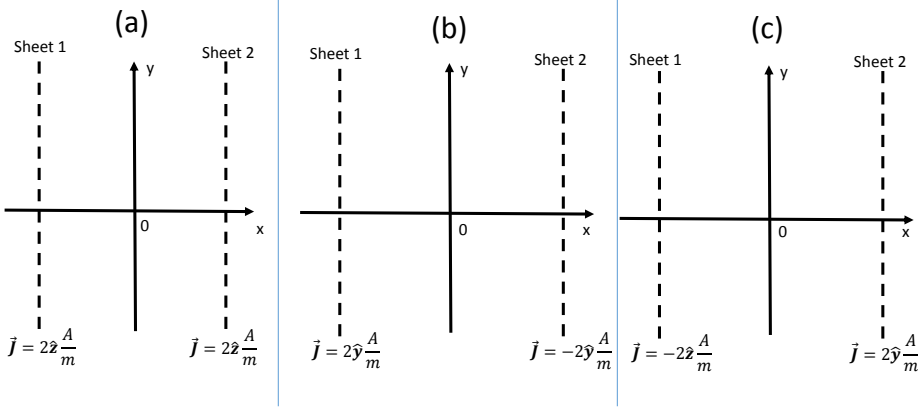
Reading Assignment: Kudeki: Lectures 12-14

1. Gauss' Law for $\mathbf{B}$ states that $\oint_S \mathbf{B} \cdot \mathbf{dS} = 0$ over any closed surface S enclosing a volume V. Given that $\mathbf{B} = B_0(-3\hat{x} + 2\hat{y} - \frac{\pi R}{L}\hat{z})$ Wb/m^2, determine the magnetic flux through the partial cone surface shown in the following figure:



2. An infinite current sheet with a uniform current density $\mathbf{J} = J_s\hat{y}\ \frac{\text{A}}{\text{m}}$ produces magnetostatic fields $\mathbf{B}$ with $\frac{\mu_o J_s}{2}$ magnitude on both sides of the sheet and with opposing directions in consistency with the right-hand-rule and the Biot-Savart law.

   Determine the **magnetic field intensity** $\mathbf{H} = \frac{\mathbf{B}}{\mu_o}$ at origin O in the following diagrams due to a pair of current sheets with specified $\mathbf{J}$ vectors.



3. Consider an infinite slab (extending in y and x directions) of a finite width $W = 4$ m described by $-3 < z < 1$. The slab is electrically neutral but it conducts a uniform current density of $\mathbf{J} = 2\hat{y}$ A/m^2 (meaning that it contains equal densities of positive and negative charge carriers moving in opposite directions parallel to $\hat{y}$). Outside the slab, that is for $z > 1$ and $z < -3$, the charge and current densities are zero.

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a) Using the right-hand-rule and Biot-Savart law, discuss why the current slab should generate equal and opposite directed magnetic fields in $\pm\hat{x}$ directions in front of and behind the plane of symmetry of the slab.

b) Based on part (a), what is $\mathbf{B}$ on the $z = -1$ m plane? Briefly explain the reasoning behind your answer.

c) Next, make use of the integral form of Ampere's law and the deductions of parts (a) and (b), to find $B_x(z)$ in the regions outside the slab. Hint: make use of a shifted coordinate system with its origin at the center of the slab.

d) Use Ampere's law to find $B_x(z)$ at a distance z within the current slab.

e) Plot B_x as a function of x over $-5 < z < 3$. Be sure to label all relevant values of B_x and z.

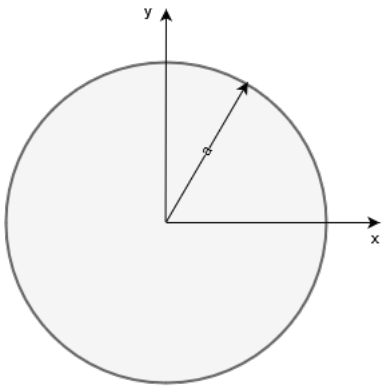


Figure A

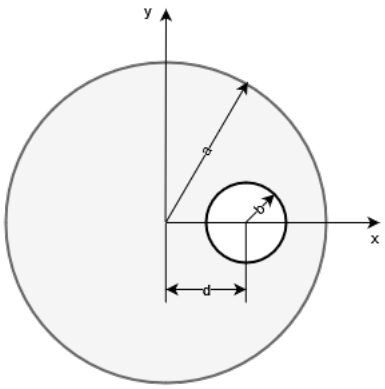


Figure B

4. Consider an infinitely long cylindrical region of current centered on the z-axis with radius a supporting a current I uniformly flowing in the $+\hat{z}$ direction, depicted in Figure A above.

a) What is the current density $\mathbf{J}$ within the cylindrical region? **Hint:** The current flows uniformly through the whole surface, so we can express its magnitude as $|\mathbf{J}| = I/A_a$, where A_a is the cross-sectional area of the cylinder in part (a).

b) Using the integral form of Ampere's law find $\mathbf{B}$ everywhere inside the cylinder.

c) Now consider the same current I flowing through the cylinder of radius a with a hole of radius b whose axis is parallel to the z-axis but offset by a distance d, depicted in Figure B. The hole carries no current. Find the current density $\mathbf{J}_c$ flowing through the region. **Hint:** See part (a) and adjust the area. Is it $\mathbf{J}_c$ larger or smaller than $\mathbf{J}$? Does this make sense physically?

d) Using the integral form of Ampere's law find $\mathbf{B}$ at the axis of the hole (at $x = d$). **Hint:** Apply superposition by splitting $\mathbf{J}_c$ into $\mathbf{J}_c = \mathbf{J}_a + \mathbf{J}_b$, where $\mathbf{J}_a$ represents a current density of magnitude $|\mathbf{J}_c|$ flowing over the full cylinder of radius a, and $\mathbf{J}_b$ a current density of magnitude $|\mathbf{J}_c|$ flowing opposite to $\mathbf{J}_a$ in the hole. Find $\mathbf{B}_a$ and $\mathbf{B}_b$ at $x = d$ resulting from $\mathbf{J}_a$ and $\mathbf{J}_b$, and use superposition to find the net flux density at that point.

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